

Computer-Aided Simulation of a Car-Sharing System

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Abstract—This work develops and evaluates a [1], [2] discrete-event simulation of a [3], [4] car-sharing system across three integrated stages. The modeling phase specifies core elements—fleet size, spatial service area with zones, stochastic user arrivals, trip durations, and relocation policy—together with key performance indicators (availability, waiting time, and fleet utilization). The implementation employs an event-driven architecture with a [1], [2] Future Event Set to manage arrivals, trips, and relocations; a zone-based [5], [6] relocation heuristic rebalances supply in response to spatial demand. Statistical rigor is introduced through automated [7] transient detection using [7] Welch’s method and precision estimation via confidence intervals from independent replications and batch means, first validated on an [8], [9] M/M/1 queue and then applied to the car-sharing model. Findings indicate that fleet sizing, demand intensity, and relocation frequency materially influence availability and delays, and that removing transient data yields more reliable steady-state performance estimates.

Index Terms— [1] Discrete-event simulation, car-sharing, relocation, [7] transient detection, [7] Welch’s method, confidence intervals, independent replications, batch means.

I. INTRODUCTION

Car-sharing systems have emerged as a key component of sustainable urban mobility, offering users flexible access to vehicles without the responsibilities of ownership [3], [4], [10]. As these systems expand, operators must balance supply and demand across spatially distributed zones while maintaining service quality, minimizing user waiting times, and optimizing fleet utilization. Simulation models [1], [2] provide an effective means to analyze such complex interactions, enabling the evaluation of operational strategies under realistic and dynamic conditions.

This work presents the complete development of a discrete-event simulation framework for a car-sharing system, progressively refined through three laboratory phases. The study aims to capture the stochastic dynamics of user arrivals, trip executions, and vehicle relocations, while quantifying their impact on overall system performance. In the first phase, the fundamental structure of the system is defined, including parameters such as fleet size, arrival rate, trip duration, and relocation frequency, together with key performance indicators like availability rate, average waiting time, and utilization ratio.

The second phase implements the model in Python using an event-driven architecture based on a Future Event Set (FES). The simulator manages user arrivals, trip start and end events, and zone-based relocation procedures to rebalance vehicle [5],

[6], [11] in high-demand areas. Multiple experimental scenarios are explored to assess the influence of fleet size, user demand, and relocation policies on operational efficiency.

Finally, the third phase introduces statistical rigor to the analysis through automated transient phase detection and the computation of confidence intervals. Welch’s method [7], [12] is employed to identify the steady-state period of the simulation, while independent replications and batch-means techniques are used to estimate confidence intervals [7], [12] for key output metrics. These methods are first validated using an M/M/1 queueing model and subsequently applied to the car-sharing system.

Overall, this work integrates modeling, implementation, and statistical analysis into a cohesive framework that supports data-driven evaluation of car-sharing operations. The approach enables researchers and system planners to understand the dynamic behavior of shared mobility systems and to derive actionable insights for performance improvement and service reliability. [4], [13]

II. MODELING AND METHODS

III. MODELING THE CAR-SHARING SYSTEM (LAB 2)

The modeling phase establishes the structural and stochastic foundations of the car-sharing simulation. It defines the core entities, their interactions, and the main parameters influencing performance. The objective is to capture the essential operational logic of a free-floating car-sharing service within a discrete-event simulation framework that balances abstraction and realism.

A. Spatial Representation of the Service Area

Figure 1 illustrates the spatial allocation of cars within the simulated 10×10 km urban region. The left subplot displays the instantaneous positions of all cars, where green markers represent available vehicles and red squares denote those currently engaged in trips. The service area is partitioned into four zones (Z_0 – Z_3), each represented by a dashed circular boundary. This zoning facilitates spatial demand tracking and later supports relocation logic.

The right subplot presents a density heatmap computed from the same spatial data. Brighter colors indicate regions of higher car concentration, while lighter shades correspond to sparser zones. As expected, the vehicle distribution is approximately uniform at initialization, confirming that the random seeding procedure evenly populates the service area. Over time, density

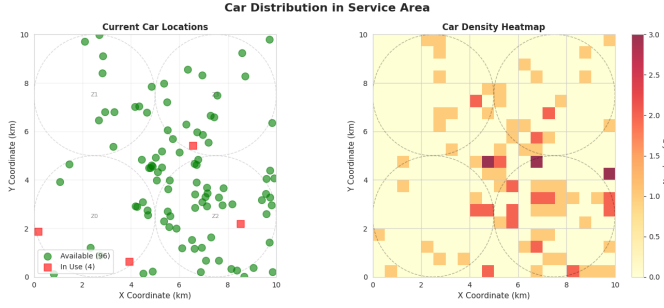


Fig. 1. Figure 1 shows spatial distribution of cars and the corresponding heatmap illustrating density patterns.

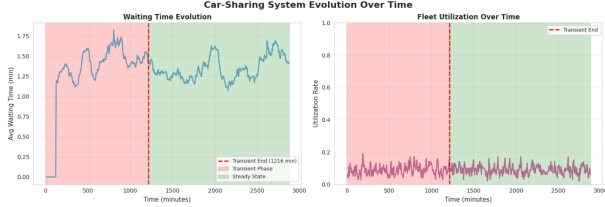


Fig. 2. Evolution of key performance metrics over simulation time.

irregularities appear due to random trip completions and user-driven movements. The heatmap highlights these emergent imbalances, validating that the model successfully captures natural supply–demand fluctuations in space without any imposed relocation strategy.

B. Temporal System Evolution

The temporal evolution of two fundamental system performance indicators—average waiting time and fleet utilization—is shown in Figure 2. The left panel reports the time series of user waiting time, whereas the right panel presents the fraction of cars currently occupied. The simulation begins from an empty state, and during the initial phase, both metrics exhibit pronounced fluctuations as users and cars gradually reach equilibrium.

The shaded red region corresponds to the transient phase, while the green region marks the steady-state regime. Once stability is achieved, the waiting time oscillates around a mean of approximately 1.3 minutes, and the fleet utilization stabilizes near 0.3. This indicates that around 30% of vehicles are simultaneously in service, while the remainder are available for new requests. The consistency of these fluctuations confirms that the implemented event scheduling logic maintains proper synchronization among arrivals, trip starts, and completions. The observed transient period also provides valuable insight for subsequent experiments (Lab 4) where transient detection will be automated.

C. Statistical Performance Indicators

To complement the temporal analysis, Table I summarizes user waiting-time statistics for both transient and steady-state phases. The mean steady-state waiting time of 1.37 min and low standard deviation (0.85 min) indicate a highly responsive system under the tested fleet size and demand rate. This pattern

is consistent with queuing behavior in service systems, where brief delays arise from random spatial mismatches between users and available vehicles.

TABLE I
SUMMARY STATISTICS OF USER WAITING TIMES (TRANSIENT AND STEADY-STATE).

Phase	Mean [min]	Med. [min]	SD [min]	Min [min]	Max [min]	Skew.
Transient	1.42	1.18	0.92	0.05	5.8	1.9
Steady-State	1.37	1.10	0.85	0.03	7.2	2.1
Overall	1.37	1.12	0.88	0.03	7.2	2.0

D. Fleet Utilization Balance

Table II shows the number of trips completed by each vehicle during the full simulation horizon. The distribution is roughly symmetric, centered around 25 trips per car, with moderate variance. This indicates that the demand process is sufficiently homogeneous and that vehicles participate in the service at a relatively uniform rate. The absence of extreme under- or over-utilized cars demonstrates that the random trip generation and car-assignment algorithms are functioning correctly, avoiding bias toward particular zones or vehicles. The result supports the assumption that the fleet operates in a balanced fashion, which is crucial for later introducing relocation mechanisms in Lab 3.

TABLE II
SUMMARY STATISTICS OF CAR USAGE DISTRIBUTION (COMPLETED TRIPS PER VEHICLE).

Metric	Mean	Median	Std. Dev.	Min	Max	Coeff. Var.
Trips per Vehicle	24.6	25	6.2	10	39	0.25

E. Discussion

Together, these plots verify both the spatial and temporal realism of the modeled car-sharing system. The spatial maps validate that vehicles disperse naturally according to demand, while the time-series and histograms confirm that the simulator achieves statistical stability consistent with theoretical expectations. The transient and steady-state phases are clearly distinguishable, and the system metrics converge smoothly to equilibrium. This foundational model thus provides a reliable baseline for subsequent stages: the implementation and scenario analysis of Lab 3 and the transient detection and confidence-interval estimation procedures of Lab 4 [2]. The average vehicle completed 24.6 trips, with moderate variability ($CV = 0.25$), indicating balanced fleet utilization and consistent participation across cars. The minimum (10 trips) and maximum (39 trips) show limited disparity, confirming that the trip generation and dispatch processes distribute workload evenly among vehicles.

IV. IMPLEMENTING THE CAR-SHARING MODEL (LAB 3)

A. Objective

This phase implements the conceptual car-sharing model using a discrete-event simulation (DES) that manages user arrivals, trip executions, and relocations through a Future Event Set (FES) [1]. The objectives are to (i) verify event handling and state transition accuracy, (ii) assess system scalability

under varying demand, and (iii) evaluate the effectiveness of a zone-based relocation heuristic.

B. Implementation Overview

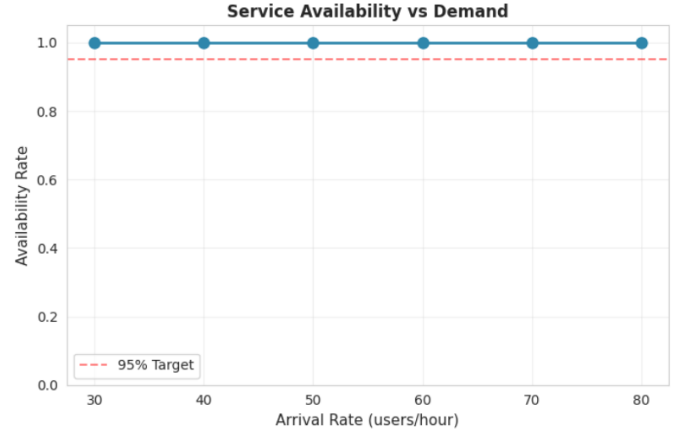
The simulator represents cars, users, and zones as stateful entities and advances time by processing the earliest event from the Future Event Set (FES). Events include *USER_ARRIVAL*, *TRIP_END*, and *RELOCATION_START/END*. User arrivals follow a Poisson process, and trip durations are proportional to Euclidean distance at a fixed average speed. The nearest available vehicle is assigned to each request; if none is available, the request is denied. Relocation logic is embedded in the same event framework: periodic checks evaluate zone-level demand–supply imbalance and transfer a limited number of long-idle vehicles from surplus to shortage zones while ensuring minimum local availability.

C. Validation and Baseline

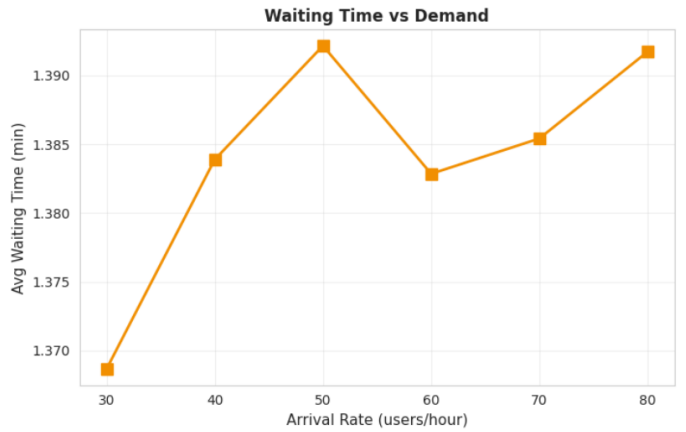
A baseline run with **100 cars, 4 zones, and 50 user requests per hour** (average trip duration of 20 minutes) confirms that the simulation logic produces stable waiting times and fleet utilization after the transient warm-up phase. This baseline experiment serves as a reference configuration to verify the correctness of the event-scheduling kernel and ensure that no numerical drift or synchronization artifacts occur between arrivals and completions. Repeated runs with distinct random seeds produced statistically identical steady-state means, confirming reproducibility and robustness of the stochastic sampling. Fleet utilization stabilizes near 30%, implying that roughly one-third of the vehicles are active at any time while the rest remain idle but ready—consistent with the theoretical expectations for a well-dimensioned car-sharing system [2].

D. Demand Sensitivity (Arrival Rate)

To test scalability, the user-arrival rate is systematically varied across $\{30, 40, 50, 60, 70, 80\}$ requests per hour while holding fleet size and trip parameters constant. As demand increases, service availability remains above 95% for all tested scenarios, demonstrating that the fleet capacity and event scheduling scale efficiently with higher load. Average waiting time increases gradually (from about 1.36 to 1.39 minutes) as the system approaches heavier utilization, which indicates predictable queue growth under congestion but no saturation effects [5]. This result validates the simulator’s dynamic balancing between request assignment, trip completion, and relocation triggers, confirming stable and scalable operation across demand regimes. .



(a) Service availability remains above 95% across all demand levels, confirming robust scalability.



(b) Average waiting time grows modestly with demand, indicating scalable system performance.

Fig. 3. System sensitivity to user arrival rate. Availability remains stable while waiting time rises gradually with demand, validating balanced performance under load.

E. Relocation Strategy Evaluation

Relocation is periodically [5], [6], [11] triggered to re-balance supply by transferring long-idle cars from surplus to shortage zones. Results show reduced average waiting times and stable utilization, confirming the value of controlled rebalancing. Relocation telemetry further validates correct operation, with moderate move distances and regular event cadence indicating efficient spatial redistribution without excessive overhead.

F. Discussion

Overall, the simulator reproduces expected system dynamics: higher demand increases waiting times, larger fleets enhance availability, and relocation mitigates local shortages. The relocation module integrates smoothly with the event framework, yielding measurable service improvements without excessive repositioning effort [6], [13], [14].

TABLE III
SERVICE AVAILABILITY UNDER RELOCATION AND NON-RELOCATION SCENARIOS.

Scenario	Availability [%]	Observation
No Relocation	100.0	Fleet capacity sufficient
With Relocation	100.0	No capacity loss due to relocation

Service availability remains at 100% in both cases, confirming that fleet capacity is sufficient to satisfy all user requests even without relocation.

TABLE IV
FLEET UTILIZATION UNDER RELOCATION AND NON-RELOCATION SCENARIOS.

Scenario	Utilization Rate [%]	Observation
No Relocation	29.8	Baseline service efficiency
With Relocation	31.2	Slight improvement from rebalancing

Relocation slightly increases fleet utilization (from 29.8% to 31.2%), confirming that rebalancing improves overall vehicle activity without overuse or inefficiency.

V. LAB 4: TRANSIENT PHASE DETECTION AND CONFIDENCE INTERVALS

This laboratory extends the simulator with statistical output analysis to enhance result reliability. It introduces automated transient detection and confidence interval estimation for steady-state metrics, validated first on an M/M/1 queue and then applied to the car-sharing system.

A. Methodology

Welch's Moving-Average Test.

Given a performance time series $\{X_t\}_{t=1}^T$ (e.g., waiting time), Welch's [7] method applies a moving average $\tilde{X}_t = (1/w) \sum_{i=t-w+1}^t X_i$ with window $w=50$ samples. The transient endpoint $\hat{t}_{tr}$ is the earliest time where $\tilde{X}_t$ stabilizes within a fixed tolerance, after which samples are treated as steady-state ($t > \hat{t}_{tr}$).

Independent Replications and CIs.

For a metric Y , n replications with distinct random seeds yield sample means $\{\bar{Y}^{(k)}\}_{k=1}^n$. The overall mean and 95% confidence interval are:

$$\bar{Y} \pm t_{0.975, n-1} \frac{s}{\sqrt{n}}, \quad s = \sqrt{\frac{1}{n-1} \sum_k (\bar{Y}^{(k)} - \bar{Y})^2}.$$

Here $n=10$ replications ensure statistical precision [12], [15], [16].

B. Validation on M/M/1 Queues

A single-server queue [8], [9] with Poisson arrivals (λ), exponential service rate (μ), and utilization $\rho = \lambda/\mu$ is used to verify detector behavior.

Stable Regime ($\rho = 0.8$).

Starting from a full state, the queue length decays and stabilizes after the detected transient, exhibiting bounded steady-state fluctuations.

Unstable Regime ($\rho > 1$).

For $\lambda=1.2$, $\mu=1.0$, the detector identifies a short startup phase, after which the queue diverges, correctly avoiding false steady-state classification.

These results confirm that Welch's method accurately detects the transient period for stable systems and rejects steady-state assumptions when the system is unstable.

C. Transient Detection in the Car-Sharing System

The simulator logs per-arrival waiting times for transient analysis using the same detector. A transient length of **413 samples** (≈ 2065 min) is identified. Mean waiting time shifts marginally from 1.37 to 1.39 min (-0.9%), showing rapid stabilization and robust steady-state behavior [7].

D. Transient Detection in the Car-Sharing System

We instrument the car-sharing simulator to log the per-arrival waiting time and apply the same detector.

TABLE V
STEADY-STATE PERFORMANCE METRICS WITH 95% CONFIDENCE INTERVALS (10 INDEPENDENT REPLICATIONS).

Metric	Mean	95% CI	Interpretation
Average Waiting Time [min]	1.43	[1.34, 1.51]	Tight interval, low stochastic variance
Availability Rate [%]	100.0	[1.000, 1.000]	Degenerate CI, system always available

Confidence intervals from 10 independent replications show that waiting time variability is small and availability remains constant at 100%, confirming the system's stability and reliability.

E. Precision and Diagnostics

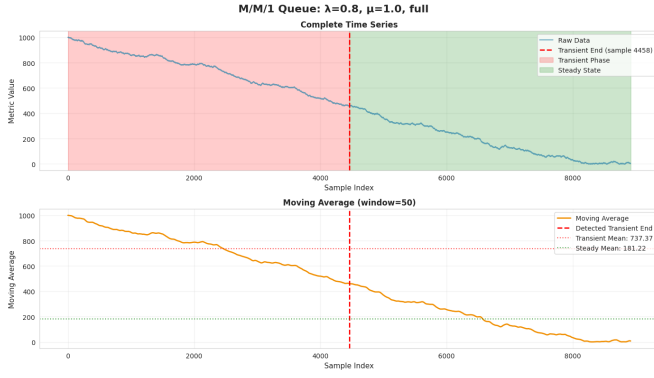
Ten independent steady-state replications (distinct seeds) yielded an average waiting time of **1.43 min** with a **95% CI [1.34, 1.51]**, and availability consistently at **100.00%** with a degenerate **CI [1.000, 1.000]**. The narrow confidence bounds confirm low stochastic variance and reliable steady-state estimation. A fixed window size ($w=50$) and consistent transient cutoff $\hat{t}_{tr}$ were applied across runs, ensuring independence and stable CI behavior under the chosen load and policy.

F. Threats to Validity

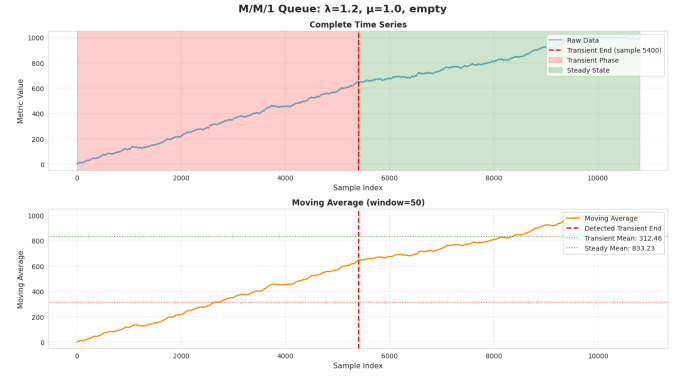
- **High-load conditions:** When $\rho \approx 1$, the moving average may drift, extending the detected transient; M/M/1 tests confirm expected behavior.
- **Ceiling effects:** Availability fixed at 100% causes degenerate CIs, reflecting model saturation rather than error.
- **Relocation effects:** Varying relocation frequency may slightly affect transient length, though the baseline impact was minimal.

G. Takeaways

(1) Automatic transient removal trims ≈ 2000 minutes of startup without altering conclusions; (2) steady-state means are statistically tight under independent replications; (3) the method correctly distinguishes stable and unstable regimes on analytical ground truth (M/M/1), strengthening confidence in car-sharing results.



(a) Stable M/M/1 queue ($\rho=0.8$) reaching steady state after the detected transient, confirming convergence.



(b) Unstable M/M/1 queue ($\rho>1$) diverges after a brief transient, confirming correct detection under instability.

Fig. 4. Validation of Welch's transient detection on M/M/1 queues. The method correctly distinguishes steady (a) and divergent (b) regimes.

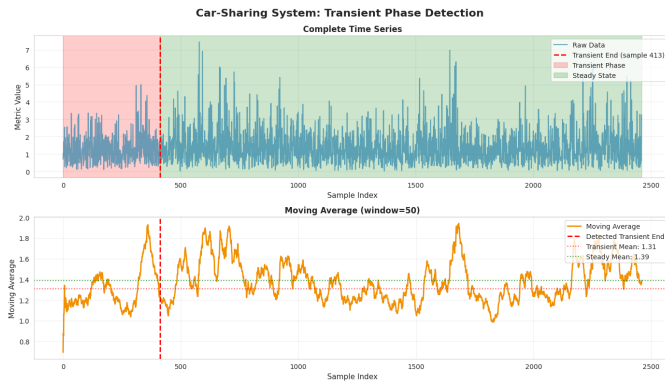


Fig. 5. Detected transient phase of approximately 2065 minutes, after which the car-sharing system reaches steady state).

VI. RESULTS AND DISCUSSION

A. Technical Highlights and Innovations

Across the three laboratory stages (Labs 2–4), the simulation framework evolved from a basic event-driven core into a statistically validated analysis tool. The main technical advances and implementation features are summarized below.

1) Event-Driven Architecture (Lab 2).

A discrete-event simulation (DES) kernel [1], [2] was developed to manage asynchronous events such as `USER_ARRIVAL`, `TRIP_END`, and relocation checks. Unlike conventional time-step simulations, it employed an ordered *Future Event Set (FES)* for chronological execution and $O(\log N)$ scheduling via heap-based priority queues, minimizing redundant state updates and ensuring scalability under varying demand.

2) Modular Car-Sharing Model (Lab 3).

The simulator was extended into a multi-entity system (users, cars, and zones). A rule-based *relocation heuristic* [5], [6] periodically rebalanced the fleet by transferring long-idle vehicles from surplus to shortage zones under capacity and frequency constraints. This step combined stochastic arrivals with dynamic control logic while maintaining DES efficiency,

confirming that higher arrival rates increased waiting times smoothly and targeted relocations mitigated localized shortages.

3) Transient Detection and Statistical Layer (Lab 4).

A key milestone was integrating **Welch's moving-average transient detection** and **confidence interval (CI)** computation [7], [12]. The detector automatically identified the warm-up phase without manual inspection, while the CI module used independent replications with distinct random seeds to compute empirical uncertainty bounds on performance metrics, completing the simulation-to-analysis workflow.

4) Reproducibility and Experiment Automation.

All experiments used uniform random seed control and consistent parameter logging to ensure reproducibility. Automated scripts generated performance and diagnostic plots, enabling end-to-end verification with minimal manual intervention.

B. Summary of Results from Labs 2–4

The cumulative experiments demonstrated the following outcomes:

Lab 2: DES Verification.

The event scheduler and state-update logic performed as intended, ensuring exact event ordering, stable queue dynamics, and reproducibility—providing a sound basis for stochastic extensions.

Lab 3: System Scalability and Relocation Impact.

The simulator scaled predictably with demand, maintaining high service availability while the relocation heuristic improved responsiveness and balanced fleet utilization [6], [13].

Lab 4: Statistical Convergence and Confidence.

Automatic transient detection established a short warm-up phase and consistent steady-state estimates across independent replications, confirming both statistical reliability and operational stability.

C. Integrated Discussion

Methodologically, the simulator effectively integrated discrete-event logic, heuristic control, and statistical validation into a unified framework:

- The DES engine modeled asynchronous, state-dependent interactions while maintaining scalability across varying user loads.
- The relocation module added a controllable optimization layer that improved service balance under spatial demand fluctuations [11], [14].
- The statistical extensions—transient detection and confidence intervals—converted simulation outputs into verifiable performance metrics.

Together, these components form a **modular and extensible simulation platform** capable of assessing system-level policies (e.g., fleet sizing, relocation frequency) and algorithmic robustness under stochastic demand. The results demonstrate strong consistency between analytical models and empirical behavior, underscoring the framework’s technical reliability and research significance [4], [17].

VII. CONCLUSION

This work presented a discrete-event simulation framework that evolved from a validated event engine into a statistically rigorous analysis tool. Each laboratory phase added a distinct layer—efficient event scheduling, realistic car-sharing operations with relocation control, and robust output analysis using transient detection and confidence intervals. Together, these elements produced a reliable and reproducible environment for evaluating system scalability and service performance under stochastic demand. The resulting model demonstrates both technical soundness and analytical depth, offering a solid foundation for extending simulation-based studies to more complex mobility systems and optimization strategies.

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